

Original Research Article

Molecular characterization of methicillin-resistant *Staphylococcus aureus*: Dissemination of multidrug-resistant community-associated MRSA and emergence of LA-MRSA, in a healthcare setting

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ABSTRACT

Background: Methicillin-resistant *Staphylococcus aureus* (MRSA) continues to be a significant cause of healthcare-associated infections (HAIs). In this study, we aimed to characterize the MRSA isolates obtained from HAIs.

Methods: A total of 200 clinical and 13 nasal MRSA isolates were collected and tested. The samples were analyzed for SCCmec typing by using multiplex PCR. Microtiter for biofilm formation were performed and molecular typing for the samples were performed for spa and agr typing.

Results: The isolates showed 100 % sensitivity to vancomycin and linezolid, while 92.5 % were multidrug-resistant. Strong biofilm-forming ability was observed in 47 % of the isolates. SCC mec typing identified 52.5 % of the isolates as classical hospital-associated MRSA or HA-MRSA (SCC mec type III), 23.5 % as community-associated MRSA or CA-MRSA (type IV and V), and 16.5 % as non-typeable, with 7.5 % having multiple SCCmec types.

Conclusion: Comparison of HA and CA-MRSA traits revealed that both groups had multidrug resistance, but HA-MRSA was distinguished by its strong capacity for biofilm formation, whereas CA-MRSA was marked by a high count of toxin gene. Our study, to the best of our awareness, documents the presence of LA-MRSA (SCCmec V-t127-agr III) causing HAIs in Indian patients for the first time.

1. Introduction

Methicillin resistant *Staphylococcus aureus* (MRSA) is known to cause serious infections associated with increased mortality, extended hospital stays, second-line drugs for treatment and substantial economic burden on healthcare systems and patients [1]. With a propensity to cause severe infections because of their multiple virulence factors and multidrug resistance [2], they are responsible for a considerable amount of healthcare-associated infections (HAIs), complicating the treatment and recovery [3]. Earlier, MRSA was known to cause infections only in the hospitalized patients with risk factors for infection. These traditional hospital-associated MRSA (HA-MRSA) are primarily responsible for severe HAIs. But, in 1990, the epidemiology of MRSA changed with the advent of community-associated MRSA (CA-MRSA), causing infection in the community setting in healthy individuals without any risk factors. HA and CA-MRSA can be distinguished by their unique molecular characteristics and drug resistance patterns. In recent years, continuous infiltration of CA-MRSA from the community into the hospitals has been

noted, posing a severe threat to vulnerable hospitalized patients [4].

Infection control is an effective measure to target and stop the transmission of these infections in hospitals. Molecular typing of bacteria is becoming inevitable in infection control because of better strain discrimination, in-depth knowledge of transmission and distribution of different clones circulating in the hospital setting [5]. Molecular typing helps track and monitor the spread of these clones and study their evolution to adapt to the hospital environment [6].

The research was conducted at the National Institute of Mental Health and Neurosciences (NIMHANS) in Bengaluru, India, which is a tertiary referral neuro-care center that serves patients from all over the country. At our hospital, MRSA infections are a frequent cause of surgical site infections and central nervous system infections. To better understand these isolates and improve our infection control program, we established a molecular surveillance setup and conducted molecular epidemiology and genetic diversity studies of MRSA causing healthcare-associated infections (HAIs). This study aimed to monitor the epidemiology of MRSA, determine if community-associated MRSA (CA-MRSA)

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has spread to our hospital, and compare the prevalence and virulence profile of HA-MRSA and CA-MRSA causing HAIs.

2. Materials and methods

2.1. MRSA collection & identification

MRSA was isolated from various clinical specimens obtained from patients epidemiologically defined to have acquired HAIs was collected from January 2015 to December 2017. The study was approved by the Institutional Ethics committee. Samples were inoculated in blood agar and MacConkey agar plates. The plates were incubated at 37 °C and observed for growth for 48 h. Blood was collected in blood culture bottles and incubated in automated blood culture systems, monitored for 5 (BD FX40 system) or 7 days (BacT/ALERT) for any bacterial growth. The MRSA isolates were identified by MALDI-TOF. Methicillin resistance was detected using 30 µg cefoxitin disc on Mueller Hinton agar.

2.2. Identification of HA and CA- MRSA by SCCmec typing

SCCmec typing was carried out using a multiplex PCR [7] with unique and specific primers, which aids in the classification of the MRSA isolates as HA-MRSA (type I, II, III) and CA-MRSA (type IV a, b, c, d, and type V). MecA was included as the internal control.

2.3. Antimicrobial susceptibility testing

MRSA isolates were subjected to Kirby Bauer disk diffusion method to determine the susceptibility pattern to different antibiotics and interpreted according to Clinical and Laboratory Standards Institute guidelines. All the discs and E-test strips for vancomycin were purchased from HiMedia Laboratories, India.

2.4. Microtiter assay for biofilm formation

The MRSA isolates ability to form biofilm was determined by the microtitre plate method. Briefly, overnight culture of the isolates were added to Brain heart infusion broth with 1 % Glucose, in respective wells of a microtitre well plate, incubated for 24 h at 37 °C. Following incubation, the wells were washed, fixed, stained and the optical density (OD) was measured at 570 nm. The isolates were categorized as strong, moderate, weak or non-biofilm formers based on the cut-off OD value determined from the OD of the negative control.

2.5. Molecular typing of MRSA isolates

The isolates were *spa* typed using previously described primers [8], which involve the PCR amplification and sequencing of the highly variable X region in the *spa* gene. The sequences were analyzed using the free software DNA Gear [9] and *spa* types were assigned to the isolates. The *agr* groups were determined by a multiplex PCR, including a pan forward primer and reverse primers specific for *agr* groups I to IV [10]. For toxin gene profiling, all the MRSA isolates were subjected to PCR amplification of 12 toxin genes encoding pore-forming toxins, superantigens and enterotoxins: *pvl*, *hla*, *tst*, *sea*, *seb*, *sec*, *seg*, *seh*, *sei*, *sem*, *sen* *seo* [11–13].

2.6. Statistical analysis

The data was analyzed using SPSS version 11.0 software. Categorical variables were represented in percentage and continuous variables were given in Mean ± SD. Chi-square test was used to analyze the categorical variable. P-value <0.05 was considered as significant.

3. Results

The isolates showing Gram-positive cocci in clusters, positive for catalase and coagulase test (tube coagulase test) and resistance to cefoxitin (a zone diameter of ≤21 mm) were considered as MRSA. A total of 200 isolates were identified as MRSA from different clinical specimens. The 13 nasal MRSA screening isolates obtained from the patients were also included for comparison with the corresponding clinical isolate. The 200 MRSA isolates were from 131 (65.5 %) male and 69 (34.5 %) female patients. There was a wide distribution of age ranging from 4 months to 84 years, with a mean age of 39.4 ± 17.5 years. The most common clinical sample was wound swab (49.5 %), followed by tracheal aspirate (25 %), and CSF (8 %) (Table 1).

3.1. Identification of HA and CA-MRSA by SCCmec typing

Only 76 % of the isolates could be grouped as HA/CA-MRSA based on SCCmec typing, which revealed that 105 isolates belonged to SCCmec type III (52.5 %), i.e. classical HA-MRSA, and 47(23.5 %) isolates belonged to classical CA-MRSA (SCCmec type IV/V), i.e. type IV in 22 isolates (11 %) and type V in 25 isolates (12.5 %). Of the remaining 24 % of the isolates, 16.5 % were non-typeable and 7.5 % of the isolates had multiple SCCmec types like combinations of I + III (3 %), III + V (0.5 %) and IV + V (4 %).

3.2. Antibiotic susceptibility

It was observed that all the 200 isolates (100 %) were sensitive to vancomycin and linezolid. The MIC values for vancomycin were in the range of 0.125–2 µg/ml higher rates of resistance were noted to

Table 1

List of the clinical specimen, patient location and associated risk factors, n = 200.

Characteristics	Number of isolates (%)
Clinical specimen	
Wound swab	99 (49.5 %)
Tracheal secretion	50 (25 %)
CSF	16 (8 %)
Blood	10 (5 %)
Shunt tips	7 (3.5 %)
CVP tips	6 (3 %)
Brain abscess pus	6 (3 %)
Sinus mucosa	3 (1.5 %)
Subgaleal pus	2(1 %)
Tissue	1 (0.5 %)
Location	
Casualty	49 (24.5 %)
ICUs	42 (21 %)
Surgery ward	31 (15.5 %)
Head injury ward	27 (13.5 %)
Step down ward	16 (8 %)
Outpatient department	15 (7.5 %)
Stroke ward	6 (3 %)
Neuro-rehabilitation ward	5 (2.5 %)
Other wards	9 (4.5 %)
Associated risk factors	
Surgery	154 (77 %)
Endotracheal intubation	96 (48 %)
ICU stay	73 (36.5 %)
Previous hospitalization	56 (28 %)
Nasal MRSA colonization	13 (6.5 %)
Underlying conditions:	
Malignancy	20 (10 %)
Diabetes mellitus	20 (10 %)
Antibiotic exposure:	
Cephalosporins	157 (78.5 %)
Aminoglycosides	68 (34 %)
Quinolones	27 (13.5 %)

antibiotics like ciprofloxacin (97.5 %), gentamicin (93 %) and erythromycin (75 %). Multidrug resistance, i.e., (resistance to ≥ 3 different antimicrobial classes), was noted in 92.5 % of the isolates.

It has also been noted that multidrug resistance was observed in 99 % of the HA-MRSA and 91.5 % of the CA-MRSA isolates (Fig. 1).

3.3. Biofilm production

Ninety-four (47 %) isolates were able to form strong biofilms; 72 (36 %) and 28 (14 %) isolates were moderate and weak biofilm producers, respectively. Among the 200 isolates, six (3 %) were non-biofilm producers. The HA-MRSA isolates (55.2 %) were significantly associated with the strong biofilm-forming ability than CA-MRSA (31.9 %) (Chi-square test, $p = 0.013$).

3.4. Molecular typing

3.4.1. *Spa* typing

A total of 27 *spa* types were noted. The most predominant types were t852 (35 %), t030 (27 %), t037 (14.5 %) and t657 (4.5 %), present in 81 % of the isolates. There were many sporadic types seen only in 1 or 2 isolates. The account of different *spa* types and their frequency are listed in Table 2.

3.4.2. *agr* typing

The predominant group was *agr* I, detected in 179 isolates (89.5 %), whereas *agr* II and III were present in 15 (7.5 %) and 5 (2.5 %), respectively. One isolate (0.5 %) was non-typeable. *agr* IV was not detected.

Regarding *agr* types, all the HA-MRSA had only *agr*I, whereas CA-MRSA had all three *agr* groups.

3.4.3. Toxin gene profiling

It was noticed that 93 % of the isolates were positive for at least one toxin gene, with *hla* being the predominant gene present in 71.5 % of the isolates, followed by the *sea* in 55.5 %. Nearly 47 % of the isolates were positive for the *pvl* toxin gene. Presence of complete enterotoxin gene cluster-*egc* cluster (*seg*, *sei*, *sem*, *sen*, *seo*) was observed in 42 % (Fig. 2).

The presence of an array of toxin genes was significantly noted in CA-MRSA compared to HA-MRSA. For example, a high number of toxin genes (≥ 5 toxins) were associated with CA-MRSA (83 %) rather than HA-MRSA (25.7 %). The *pvl* gene was present in 22.9 % of HA-MRSA isolates, but a significant association was noted with CA-MRSA (87.2 %). Likewise, the *egc* cluster was significantly associated with CA-MRSA than HA-MRSA shown in Table 3.

3.5. Comparison of nasal and clinical MRSA strain

The 13 nasal MRSA isolates were compared with the corresponding clinical isolates of the patients. Ten nasal isolates (77 %) were similar to their clinical isolates in terms of SCCmec type, *agr* type, *spa* type, virulence gene profile and biofilm production.

Table 2

spa types of MRSA isolates, $n = 200$.

<i>spa</i> types	Number of isolates (%)
t852	70 (35)
t030	54 (27)
t037	29 (14.5)
t657	9 (4.5)
t005,t127,t309, t425	3 (1.5) each type
t021,t064,t16138,t2647,t3387,t3841,t4422	2 (1) each type
t1223,t13180,t15984,t311,t345,t4285,t4326, t442,t5128, t5335,t902,t9105	1 (0.5) each type

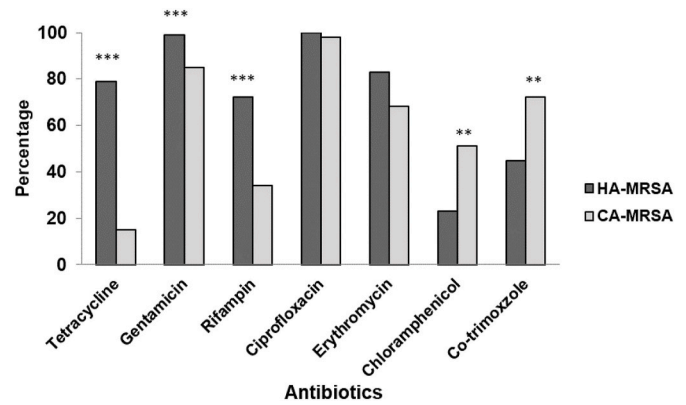


Fig. 1. Antibiotic resistance pattern of HA-MRSA ($n = 105$) and CA-MRSA ($n = 47$)

** denotes p value < 0.01 , *** denotes p value < 0.001 .

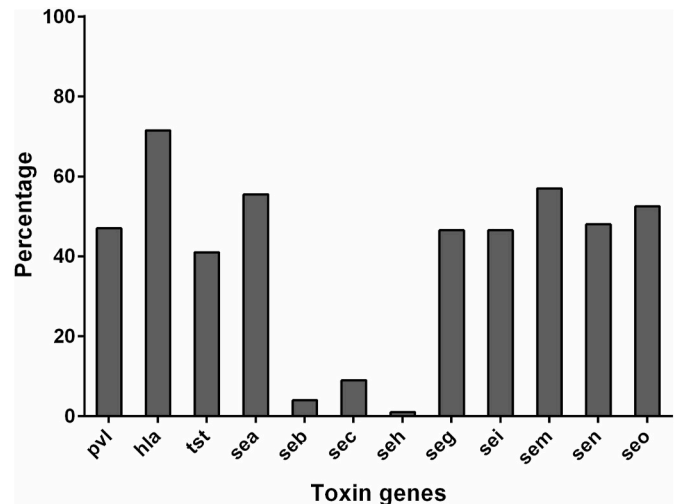


Fig. 2. Distribution of toxin genes among the MRSA isolates, $n = 200$.

3.6. Predominant genotypes

The most common HA-MRSA genotypes circulating in the hospital were SCCmec III- *spa* type t030-*agr* I (27 %) and SCCmec III- *spa* type t037-*agr* I (11.5 %) whereas SCCmec IV- *spa* type t852 - *agr* I (6 %) and SCCmec V- *spa* type t657 - *agr* II (4.5 %) were the predominant CA-MRSA genotypes.

4. Discussion

During the study period, MRSA comprised 40 % of total *S. aureus* cases in our hospital, predominantly affecting males aged 21–40. Most male patients had sustained head injuries from road traffic accidents, followed by emergency surgeries that led to MRSA infections during hospitalization. Furthermore, 50.5 % of the isolates were from SSIs. Patel et al. (2017) reported an average of 37.9 % MRSA infections among total *S. aureus* SSIs [14].

In this study, 7.5 % of the isolates contained multiple SCCmec types, indicating the co-existence of different SCCmec elements, a common occurrence in MRCoNS, which serves as a SCCmec reservoir [15]. Though rarely observed, Bhutia et al. (2015) and Nagasundaram et al. (2019) found multiple SCCmec types in 73.5 % and 59 % of their MRSA isolates, respectively. These strains often exhibit composite SCCmec elements, integrating new elements with existing ones, resulting in

Table 3 (continued)

SCCmec type (n)	agr type (n)	spa type (n)	Toxin genes (n)
NT (33)	III (2)	t309 (3)	<i>pvl, hla, sem, sen, seo</i> (1)
			<i>pvl, hla, seg, sei, sem, sen, seo</i> (3)
			<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (4)
			<i>pvl, hla, tst, sea, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, sem, sen, seo</i> (1)
		t005 (3)	<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (2)
			<i>pvl, tst, seg, sei</i> (1)
		t3841 (1)	<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (2)
			None (1)
			<i>pvl, sea, sen, seo</i> (1)
	NT (1)	t021 (2)	<i>pvl, hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, sem, sen, seo</i> (1)
		t5335 (1)	None (1)
			<i>sei, sem</i> (1)
			<i>pvl, sem, sen, seo</i> (1)
	I (33)	t852 (27)	<i>pvl, seg, sei, sem, seo</i> (1)
			<i>pvl, hla, seg, sei, sem, seo</i> (1)
			<i>hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, seg, sei, sem, sen, seo</i> (4)
			<i>pvl, tst, seg, sei, sem, sen, seo</i> (3)
			<i>pvl, hla, tst, seg, sei, sem, seo</i> (1)
			<i>pvl, hla, seg, sei, sem, sen, seo</i> (7)
			<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (2)
			<i>pvl, hla, sea, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, tst, sea, seg, sei, sem, sen, seo</i> (3)
Multiple type (15)	I + III (6)	t16138 (2)	<i>pvl, hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (1)
			<i>hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl</i> (1)
			<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (1)
		t902 (1)	<i>hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl</i> (1)
			<i>pvl, hla, seg, sei, sem, sen, seo</i> (1)
			<i>pvl</i> (1)
I + III (6)	I (6)	t037 (6)	<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (1)
			<i>hla, sea</i> (2)
			<i>hla, tst, sea</i> (3)
			<i>pvl, hla, tst, sea, sem</i> (1)
			<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (2)
IV + V (8)	I (8)	t852 (8)	<i>pvl, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, seg, sei, sem, sen, seo</i> (3)
			<i>pvl, hla, sea, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, tst, seg, sei, sem, sen, seo</i> (1)
			<i>pvl, hla, tst, sea, seg, sei, sem, sen, seo</i> (1)
III + V (1)	I (1)	t852 (1)	<i>pvl, hla, tst, sea, seg, sei, sem, sen, seo</i> (1)

High resistance to commonly used antibiotics like ciprofloxacin, erythromycin, and gentamicin was observed, consistent with other studies [18]. Most isolates were also resistant to tetracycline, co-trimoxazole, and rifampin, aligning with another study's findings. The rising multidrug-resistant MRSA isolates underscore the need for stringent infection control and antibiotic stewardship. Susceptibility to non-beta-lactam antibiotics typically marks CA-MRSA. Traditional HA-MRSA, benefiting from strong biofilm formation and multidrug resistance, thrive in critically ill, immunocompromised patients with life support devices, as evidenced by the high number of HA-MRSA in ICUs.

The spa typing revealed four predominant types—t852, t030, t037, t657—accounting for 81 % of isolates, followed by types like t021, t005, t425, t064, t3387, t307, commonly reported in India [19]. Notably, Spa type t127, previously unreported in Indian MRSA isolates, was found in this study. The genotype (SCCmec V-t127-agr III) is typically isolated from livestock [20]. This finding suggests the threat of livestock-associated MRSA (LA-MRSA) entering hospitals. A study by merla et al., 2022 found that 1.42 % of blood culture were LA-MRSA i.e. five isolates amongst 353 MRSA and 1 % of other clinical samples i.e. nine isolates amongst 904 MRSA [21].

other studies [22]. The emergence of CA-MRSA harboring various toxin genes is a matter of concern, as the spread of virulent CA-MRSA in the hospital is a considerable risk to the patients.

We also observed that the multiple SCCmec MRSA could be grouped as HA or CA-MRSA based on the SCCmec combination they carry, as it was evident that type I + III resembled HA-MRSA and type IV + V was similar to CA-MRSA in their characteristics. The increasing incidence of NT and multiple SCCmec type strains shows the drastic change in MRSA epidemiology and makes it difficult to classify into HA or CA-MRSA based only on SCCmec types.

Among the thirteen patients with nasal MRSA carriage (77 %), were colonized with the strain indistinguishable from their clinical strain, indicating the potentiality of the colonizing strain causing invasive infection, thus implying a considerable risk in hospitalized patients. This finding supports the concept of preoperative/admission screening and decolonization strategy in preventing infection from endogenous MRSA in susceptible individuals, which has been proven in many large-scale studies [23].

Another common clone in this study SCCmec IV- *spa* type t852 belonging to MLST ST22 (CC22), is known to have moved from community to hospitals [24]. This ST22-Scmec IV, also known as EMRSA-15, is already a well-established hospital clone and has replaced the traditional hospital strains in most European countries [25] and appears to be more virulent and highly transmissible. The other predominant clone in this study is SCCmec V- *spa* type t657, which belongs to MLST ST772 (CC1), a Bengal Bay clone carrying *pvl* gene along with multiple virulence factors and multidrug-resistance is widespread in many countries like India, Malaysia, UK, Ireland. The two CA-MRSA clones mentioned above are increasingly being reported and are considered emerging in Indian hospitals [26]. It is also predicted that these two clones may replace the traditional clones and successfully establish themselves in the hospital because of their multidrug resistance and multiple virulent toxin genes.

5. Conclusions

The study reflects on diverse MRSA clones circulating in the hospital, causing healthcare-associated infections. Characterization of MRSA isolates highlighted the existence of strong biofilm-forming, highly drug-resistant HA-MRSA; substantial infiltration and dissemination of CA-MRSA carrying potential toxin genes with multidrug resistance, and also the emergence of LA-MRSA in our hospital.

CRedit authorship contribution statement

Vijayan Priya: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **S. Nagarathna:** Writing – review & editing, Visualization, Supervision, Investigation, Formal analysis. **Kumari HB. Veena:** Writing – review & editing, Validation, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Ethics

Ethical approval were taken from the Institutional Ethics committee. NIMHANS IEC No. Item No.1., Sl.No. 1.01 (Basic Sciences).

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Declaration of competing interest

None.

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